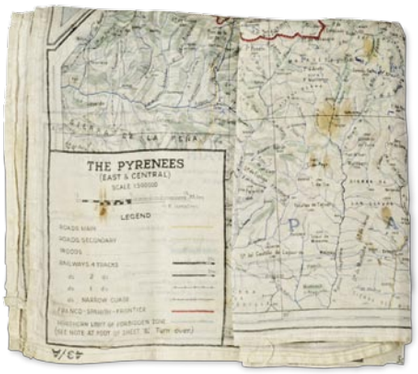
Inexperienced pilots were liable to fire at any aircraft they saw, friend or foe. Out-turning an enemy to get on his tail was the most successful tactic, but if a pilot turned his fighter too tightly he could pass out through excessive g-forces.

Attacking the mass of slower-moving Dornier, Junkers, and Heinkel bombers posed its own problems. The most effective tactic was to fly head-on towards the bombers, which could break up the formation. It was also incredibly risky and psychologically demanding. Most pilots settled for attacking bombers from the flank or rear. Bombers were relatively easy to hit but were hard to shoot down, absorbing a great deal of punishment. The fighter had to fire at close range to achieve any decisive effect, taking the chance of being himself hit by the bombers' well-trained gunners.

## THE BEST AND THE REST

When aircraft flying at over 480kph (300mph) met in aerial combat, the speed at which events occurred required a pilot to possess very special qualities. In the pre-computer age, just controlling the aircraft was a refined skill, even without the need to manoeuvre in a crowded air space and locate and fire on a target. It has been estimated that no more than one in 20 Battle of Britain pilots had the combination of flying skills, superb eyesight, instant reactions, situational awareness, and killer instinct to make a really effective fighter pilot. A small number of aces accounted for a large proportion of enemy aircraft shot down – men such as South African Adolph "Sailor" Malan, Czech pilot Josef Frantisek, and British Sergeant "Ginger" Lacey. At the other extreme, new pilots thrown into combat after around 12 hours flying time on fighter planes were quite likely to be killed on their first or second mission – especially as squadron leaders, keen to hold onto their experienced pilots, would often put the novices in the most vulnerable position at the rear of the formation. Pilot Hugh Dundas recalled being "close to panic in the bewilderment and hot fear" of his first dogfight. But at least "as the silhouette of a Messerschmitt passed by", he managed to fire his guns. There were many pilots who went through their first experience of air combat without seeing the enemy at all – everything simply happened too fast.




**Silk map**
This light, silk map was stitched into the pilot's jacket. It was needed if he was forced to bail out – in this case, over the Pyrenees.

The performance of Fighter Command improved as experience built up in daily combat. Many second-rate squadron leaders were quickly replaced, and some outstanding figures emerged, such as Peter Townsend, in command of 85 Squadron, and Malan leading 74 Squadron. As the days passed, skilled fighters spread information about successful methods of combat, so that, for example, pilots learned the importance of only firing when close in to the enemy, setting their guns to fire in a cone converging at

> "THE BEST DEFENCE OF THE COUNTRY IS THE FEAR OF THE FIGHTER ... IF WE ARE WEAK IN FIGHTER STRENGTH ... THE PRODUCTIVE CAPACITY OF THE COUNTRY WILL BE VIRTUALLY DESTROYED."
>
> **SIR HUGH DOWDING**, AIR CHIEF MARSHAL



**Hurricane squadron**
Twelve pilots scramble for their Hurricane fighters during a peacetime exercise in 1939. Each wears a seat-type parachute over his flying suit.